

Literature Survey and Critical Analysis Assignment

Deadline: April 26, 23:59

Overview

The goal of this assignment is to encourage you to explore the current literature, evaluate state-of-the-art methodologies, and critically analyze their limitations through empirical observation.

You will conduct a literature survey on a topic related to the course and experimentally evaluate several state-of-the-art methods by running their official implementations. Through these experiments, you are expected to identify the limitations and failure cases of existing approaches.

Assignment Structure

Please complete the following steps and compile your findings into a short academic report.

1. Literature Survey

Choose a specific topic from the course. Then identify and review **six research papers** related to this topic that were not covered during the lectures.

For each paper, briefly summarize:

- The main idea and contribution of the work
- The core methodology or model used
- The problem that the paper aims to solve

2. Running Existing Methods

Among the surveyed papers, select **three papers** that:

- Provide an **official code implementation**
- Report **state-of-the-art or strong benchmark results**

Set up their environments and run the inference code using your own selected example images. Include the resulting outputs in your report.

3. Failure Case Analysis

Analyze the outputs generated by these three algorithms and identify cases where they:

- Fail to produce correct results
- Produce visual artifacts or unrealistic outputs

For each failure case:

- Provide the input image
- Show the output of the algorithm
- Explain *why* you believe the method fails in that case

Your discussion should demonstrate critical thinking about the limitations of the methods.

Submission Requirements

- The report must be written using the **CVPR conference template**.
- The report must not exceed **3 pages**. References are **not included** in the page limit.
- This is an **individual assignment**. Each student must submit their own report.
- Your final submission must be a single **PDF** file.
- All **six surveyed papers must be properly cited** in a *References* section.
- Include **visual results** for both successful examples and failure cases.

Evaluation Criteria

Your report will be evaluated based on:

- Quality and relevance of the literature survey
- Correct setup and execution of the selected methods
- Depth of the failure case analysis
- Clarity and organization of the report